(5) $L_1 = \lim_{x \to 11} \frac{x^2 - 11^2}{x - 11}$ faktorkan $= x^2 - 121$

$= (x-11)(x+11)$

$= \lim_{x \to 11} (x+11) = 22$

$L_2 = \lim_{x \to 0} \frac{\sqrt{x + (k+1)^2} - (k+1)}{x}$ $k+1 = 12$

$(k+1)^2 = 144$

$= \lim_{x \to 0} \frac{(\sqrt{x+144} - 12)(\sqrt{x+144} + 12)}{x(\sqrt{x+144} + 12} = \frac{(x+144) - 144}{x} =$

$= \lim_{x \to 0} \frac{1}{\sqrt{x+144} + 12} = \frac{1}{\sqrt{144} + 12} = \frac{1}{24}$ //

$L_3 = \lim_{x \to \infty} \frac{(a+2)x^3 + b.x + 7}{x^3 - x + k}$ $a+2 = 4$ $k = 11$

$b = 8$

$= \lim_{x \to \infty} \frac{4 + \frac{8}{x^2} + \frac{7}{x^3}}{1 - \frac{1}{x^2} + \frac{11}{x^3}} = \frac{4}{1} = 4$

$L_4 = \lim_{x \to \infty} \frac{\sin((a+1)x) \cdot \tan((b+2)x)}{x^2}$ $a+1 = 3$

$b+2 = 10$

$= \lim_{x \to \infty} \frac{(3x)(10x)}{x^2} = 30$ //

$R(n) = \frac{100n + k}{n + (a+b+1)} = \frac{100n + 11}{n + (2+8+1)} = \frac{100n + 11}{n + 11}$

$\lim_{n \to \infty} R(n) = \lim_{n \to \infty} \frac{100 + \frac{11}{n}}{1 + \frac{11}{n}} = 100$ //

Dipindai dengan CamScanner